

Styling Your Code

Now that you're writing longer programs, it's a good idea to learn how to style your code consistently. Take the time to make your code as easy as possible to read. Writing easy-to-read code helps you keep track of what your programs are doing and helps others understand your code as well.

Python programmers have agreed on a number of styling conventions to ensure that everyone's code is structured in roughly the same way. Once you've learned to write clean Python code, you should be able to understand the overall structure of anyone else's Python code, as long as they follow the same guidelines. If you're hoping to become a professional programmer at some point, you should begin following these guidelines as soon as possible to develop good habits.

The Style Guide

When someone wants to make a change to the Python language, they write a *Python Enhancement Proposal (PEP)*. One of the oldest PEPs is *PEP 8*, which instructs Python programmers on how to style their code. PEP 8 is fairly lengthy, but much of it relates to more complex coding structures than what you've seen so far.

The Python style guide was written with the understanding that code is read more often than it is written. You'll write your code once and then start reading it as you begin debugging. When you add features to a program, you'll spend more time reading your code. When you share your code with other programmers, they'll read your code as well.

Given the choice between writing code that's easier to write or code that's easier to read, Python programmers will almost always encourage you to write code that's easier to read. The following guidelines will help you write clear code from the start.

Indentation

PEP 8 recommends that you use four spaces per indentation level. Using four spaces improves readability while leaving room for multiple levels of indentation on each line.

In a word processing document, people often use tabs rather than spaces to indent. This works well for word processing documents, but the Python interpreter gets confused when tabs are mixed with spaces. Every text editor provides a setting that lets you use the TAB key but then converts each tab to a set number of spaces. You should definitely use your TAB key, but also make sure your editor is set to insert spaces rather than tabs into your document.

Mixing tabs and spaces in your file can cause problems that are very difficult to diagnose. If you think you have a mix of tabs and spaces, you can convert all tabs in a file to spaces in most editors.